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Predictive process mapping for laser powder bed fusion: A review of existing analytical solutions

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ABSTRACT

One of the main challenges in the laser powder bed fusion (LPBF) process is making dense and defect-free components. These porosity defects are dependent upon the melt pool geometry and the processing conditions. Power-velocity (PV) processing maps can aid in visualizing the effects of LPBF processing variables and mapping different defect regimes such as lack-of-fusion, under-melting, balling, and keyholing. This work presents an assessment of existing analytical equations and models that provide an estimate of the melt pool geometry as a function of material properties. The melt pool equations are then combined with defect criteria to provide a quick approximation of the PV processing maps for a variety of materials. Finally, the predictions of these processing maps are compared with experimental data from the literature. The predictive processing maps can be computed quickly and can be coupled with dimensionless numbers and high-throughput (HT) experiments for validation. The present work provides a boundary framework for designing the optimal processing parameters for new metals and alloys based on existing analytical solutions.

1. Introduction

Laser powder bed fusion (LPBF), also called selective laser melting (SLM), offers a constraint-free design space to fabricate complex geometries in a single processing step [1,2]. The growing popularity of the LPBF process amplifies the need to fabricate high quality and structurally sound components with high repeatability. In the LPBF process, there are at least 100 independently controlled processing parameters [3]. The most common and widely studied parameters are laser power, scanning speed, hatch spacing, layer thickness, and laser spot size [2,4–6]. These primary variables control the amount of energy supplied per unit volume and the rate of material deposition. Other important variables such as scan strategy, build plate temperature, support structures, etc. have an effect on residual stress and heat distribution within the components [7–9]. The primary objective is controlling porosity defects, and within viable process parameter windows, microstructures can be subsequently tailored for specific properties. Identifying the optimal processing condition for a given material is a challenging and time-consuming task, mainly due to the possible range of each parameter. Often, the manufacturer's recommendation is used to select processing parameters, but for new alloy systems or property-tailoring of a manufacturer's recommendation, trial-and-error methods are often required [10,11]. To fully exploit the LPBF process, rapid identification and validation of the processing window are needed.

Power-Velocity (PV) processing maps are a powerful tool to visualize the effects of LPBF processing variables to produce high density parts. Processing maps have been systematically used by Beuth *et al.* [12] to overlay the contours of part's density, surface roughness, and build rate onto a PV map. A few other recent studies [13,14] have extended this idea and constructed processing maps for different materials like titanium, Ni-Nb, and a high entropy alloy. Depending upon the laser power and scanning speed combinations, three distinct porosity defects, lack-of-fusion, balling, and keyholing can be mapped onto the PV processing map as shown in Fig. 1a. Lack-of-fusion defects form due to insufficient melt pool overlapping and a shallow melt pool [15]. The underdeveloped melt pool or large hatch spacing results in the formation of cavities and porosities at the vicinity of the melt pool boundary as shown in Fig. 1b and d. These defects have high aspect ratios, sharp corners, and their size can range from 1 to 100 μm. Low laser power or high scanning speed can make the part vulnerable to lack-of-fusion defects [16,17]. In a study by Mukherjee *et al.* [18], a normalized heat-input parameter (a non-dimensional number) showed a linear dependence with the number of lack-of-fusion defects. Balling (or Rayleigh-

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Nomenclature

α	Laser absorptivity
A	Melt pool cross sectional area
β	Coefficient of linear expansion
C_p	Specific heat
d	Melt pool depth
d_l	Laser spot diameter
D_t	Thermal diffusivity
$\in$	Cooling rate
$\in_t$	Thermal strain
E	Elastic modulus
$\bar{E}$	Material toughness near the solidus temperature
f_s	Solid fraction of semi-solid region
Fo	Fourier number
G	Thermal gradient
H	Hatch spacing
ΔH	Latent heat
I	Moment of inertia
J	Vaporization flux
K	Thermal conductivity
Ke	Keyhole number
λ	Evaporation energy per atom
λ_{PDAS}	Primary dendritic arm spacing
l	Melt pool length
L	Layer thickness
L_c	Characteristic length
Δm	Material vaporized
M	Molecular weight
P	Laser power
P_s	Saturated vapor pressure
P^*	Chamber pressure
P_r	Recoil pressure
Pe	Peclet number
ρ	Density
r_l	Laser spot radius
R	Solidification rate/Molar gas constant/Melt pool aspect ratio
s_f	Shrinkage factor
τ	Characteristic time
T_m	Melting temperature
T_b	Boiling temperature
T_o	Build plate temperature
ΔT	Difference between peak and melting temperature
$T_{surface}$	Surface temperature
$T_{solidus}$	Solidus temperature
$T_{liquidus}$	Liquidous temperature
U_{max}	Maximum velocity of molten metal
V	Scanning speed
w	Melt pool width
X	Mole fraction

Plateau) is a melt pool instability in which a small hump forms along the printed track. The high contact angle between the spherical humps and the previously solidified layer results in porosity formation as shown in Fig. 1b and e. Balling can also affect the subsequent layer's powder packing, imparting processing defects and increasing the part's surface roughness. High laser power and high scanning speed can promote balling [19,20]. In such cases, the melt pool length is much greater than the melt pool width. Keyholing, in contrast to the lack of fusion, occurs due to excessive penetration of the fusion zone. Higher vaporization and recoil pressure result in cavity formation between the melt pool boundaries as shown in Fig. 1b and f. These defects generally have ellipsoidal or spherical geometries. Cunningham *et al.* [21] found that keyholing occurs when the keyhole front angle exceeds a critical value of 77 degrees. High laser power and low scanning speed increase the melt pool depth and thus the potential of keyhole defects [21]. Thus, control over the melt pool geometry via processing parameters is crucial for good interlayer bonding and defining an alloy's optimum printability range as represented in a PV map.

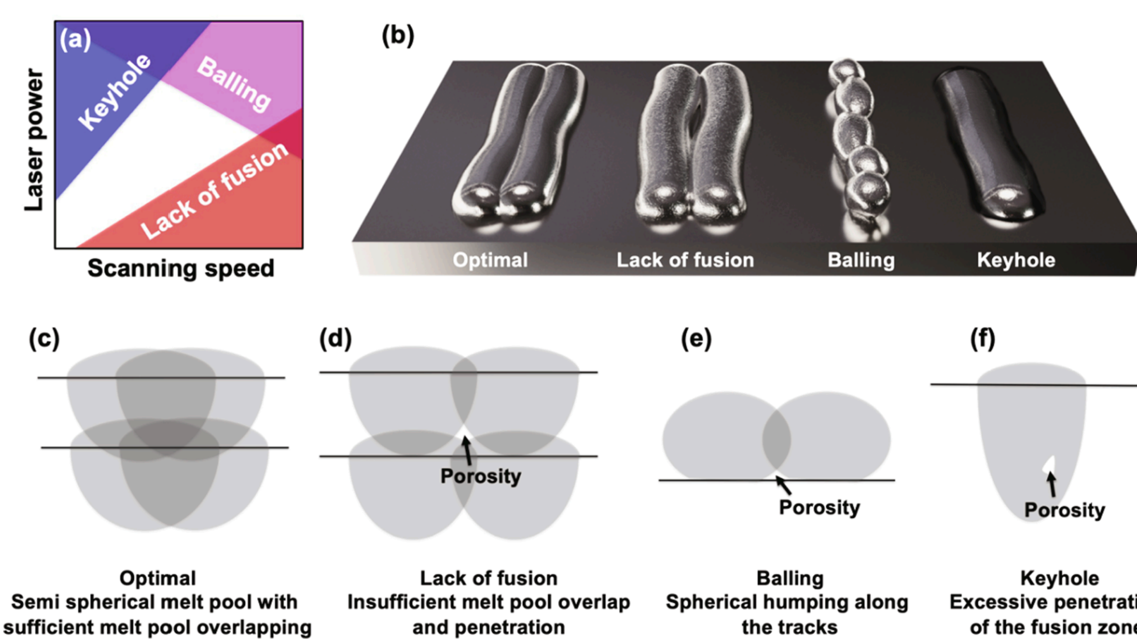

Fig. 1. (a) PV processing map for the LPBF process showing three distinct porosity defects: keyhole, balling, and lack of fusion. (b) Simplified illustration of the melt pool tracks at different processing regime of the PV maps, (c-f) simplified cross section of the melt pool tracks and site of occurrence for porosity defects.

Processing parameters combined with the material's thermophysical properties dictate the laser-metal interaction, and therefore, the melt pool geometry. Prior studies have used *in situ* x-ray and thermal imaging techniques to observe the laser-metal interaction at a high spatial and temporal resolution [22,23]. The intensity of the synchrotron x-ray images represents density variations which help in identifying melt pool morphology, tracking spatter particles, and understanding pore formation mechanisms [23–26]. In a study by Guo *et al.* [27], synchrotron x-rays were used to calculate the melt pool dimensions at different processing conditions. Usually, the melt pool morphology changes from semi-spherical (*width* ≈ 2**depth*) to deep (*depth*≫*width*) with increasing laser power. This change in melt pool morphology occurs with a simultaneous increase in melt pool width and melt pool depth [28]. In addition to synchrotron x-ray imaging, infrared cameras can be used to map the temperature distribution during melting and solidification. This technique provides information such as melt pool dimensions, peak temperature, local heating cycles, and can be applied as a feedback control system for processing parameters [29,30]. Although the resolution of images obtained from infrared cameras is less than synchrotron x-ray imaging, they are easier to acquire and analyze.

In addition to *in situ* techniques, other studies have focused on mathematical modelling techniques to elucidate the LPBF process. For example, finite element method (FEM) models have been used to calculate the heat transfer and solidification conditions, and computational fluid dynamic (CFD) models have been used to study the melt pool dynamics [5,31,32]. In a study by Lee *et al.* [33], thermal gradient and solidification rate were calculated to compare the solidification morphologies at different processing conditions for LPBF of Inconel 718. Similarly, analytical models have also been developed to predict the melt pool geometry, ideal hatch spacing, and defect boundaries [18,20,34–36]. Unlike FEM simulations, analytical models are time-efficient, computationally inexpensive, and they provide a quick approximation. However, it is difficult to accurately model the heat transfer and melt pool dynamics of the LPBF process using analytical models due to the multi-physics nature of the process which extends across time and length scales.

High-throughput (HT) experiments [37,38] have gained interest across various disciplines of discovery science. An HT approach allows to design, fabricate, characterize, and analyze multiple samples at once and rapidly generate useful data. The data obtained from the HT approach may show marginal deviations, as the rapid analysis compromises the measurement resolution. Nevertheless, the HT experiments are effective for quick interrogations [39]. In a recent study by Salzbrenner *et al.* [40], an HT technique was used to evaluate the tensile performance of additively manufactured parts. Specifically, the HT approach can be beneficial for the LPBF process because of its large processing space. In a recent study by Agrawal *et al.* [41], a combination of HT fabrication and characterization led to rapid identification of the processing bounds for 316L stainless steel. Large datasets obtained from the HT approach can also be combined with various statistical analyses and machine learning algorithms to build robust predictive models [42,43]. For example, Ren *et al.* [44] combined the HT approach and machine learning to discover new metallic glass compositions. HT experiments can also help to discover new dimensionless numbers offering predictive capability. In the study by Rankouhi *et al.* [45] a dimensionless number was discovered which showed a correlation with the part's density for different metals and alloys.

Additive manufacturing offers an opportunity to discover new alloy compositions and take advantage of its constraints-free alloy design space. For example, the accelerated discovery of new high entropy alloys [46] and bulk metallic glass compositions [44] has been demonstrated using directed energy deposition (DED) technique. On the contrary, the number of newly discovered compositions for LPBF fabrication remains limited. This is partly because different alloys need different processing strategies which pose a set of challenges unique to each alloy. For example, low laser absorptivity is an issue for fabricating pure copper, high oxygen affinity makes processing of aluminum powders challenging, and high melting points of tungsten and molybdenum require high input energy [47–49]. Moreover, a poor feedstock powder quality and non-uniform powder size distribution can add additional complexity to this problem [50]. Although the experimental and modelling studies on parameter optimization are powerful tools on a case-by-case basis, they are not universal. For example, in many of the mathematical models, a 2D model of heat transfer is applied instead of a 3D model to simplify the physical problem. Similarly, the experimental results obtained can vary from one LPBF unit to another as repeatability and certification are still an issue with AM technologies. Thus, it becomes important to critically review these studies and understand the circumstances where they offer good predictability. As a subsequent focus area, microstructural control for property tailoring is important, but establishing high-density process parameter windows is a required first step and is the focus of this study.

In this work, we aim to combine existing analytical models and porosity defect criteria to rapidly generate PV processing maps and compare them with experimental data from the literature. We especially aim to predict the PV processing window for the LPBF process as a function of hatch spacing, layer thickness, and material properties. The general theme of the paper is presented in Fig. 2. Laser processing parameters and materials properties serve as input variables to construct the LPBF processing map. These sets of input variables are either constant for a given material (like materials properties) or can be independently controlled (like laser parameters) during LPBF processing. The only exception is the laser absorption coefficient which can vary with powder size distribution. A brief discussion on the input parameters and how these variables affect the laser-metal interaction is presented in Section 2. Several design parameters and dimensionless numbers have been developed to correlate the processing parameters and describe various aspects of the LPBF process. A comparison of these commonly used design parameters along with their strength and limitations is discussed in Section 3. In Section 4, different analytical models available in the literature are presented to quickly estimate the melt pool geometry including melt pool width, depth, and length. Control over the melt pool is crucial, if porosity defects in AM components are to be avoided. The evaluation in Section 5 discusses how the melt pool geometry affects porosity and details the existing defect criteria for lack-of-fusion, undermelting, balling, and keyholing regions. In Section 6, additional analytical models and dimensionless numbers are discussed highlighting the dependence of laser-metal interaction on compositional changes, thermal strain, depression zone, solidification cracking, and as-solidified microstructure. The estimates of melt pool geometry and defect criteria were combined to construct and compare the LPBF processing maps for different materials. Finally, the processing maps predictions are then compared with existing experimental data gathered from the literature to provide an assessment of the predictive capability of these maps. These results are presented in Section 7.

This article offers three benefits: (i) a critical review of the analytical models available in the literature, (ii) an insight into different porosity defects and strategies to mitigate them, and (iii) a predictive tool that serves as a feedback loop to quickly identify the LPBF processing window.

2. Input variables

2.1. Material properties

The thermophysical properties of materials provide information about the material's behavior under the application of heat. In the LPBF process, thermophysical properties influence the laser-metal interaction and thus the melting and heat transfer [4,33,51]. The most influential thermophysical properties within the context of the LPBF process are melting temperature, density, specific heat, thermal conductivity, and thermal diffusivity. These properties are either related to

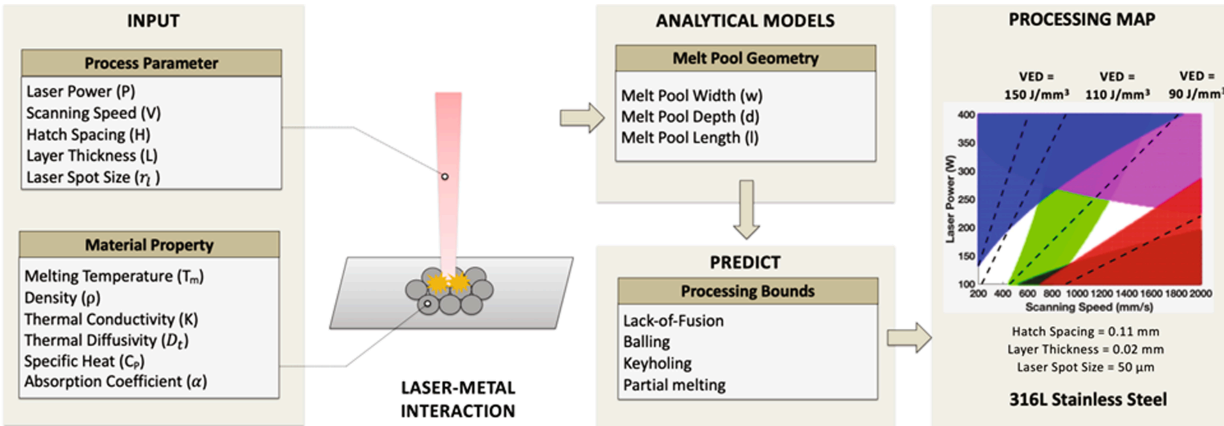

Fig. 2. The general theme of the current review paper. A set of process parameters and materials properties were selected as an input variable to predict the melt pool geometry and LPBF processing bounds of a material.

thermodynamics or heat transfer phenomena. For example, specific heat is a thermodynamic variable, while thermal conductivity is a variable affecting heat transfer.

A material's intrinsic thermophysical properties like melting temperature, density, and specific heat are constant for a given crystal structure. In another words, their value does not vary with powder particle size or by the mass of powder being exposed to the laser beam. Material density is defined by atomic fraction and location of atoms in a unit cell. The melting point is related to the bond strength while the specific heat is dictated by the lattice vibrations. Typically, a material having a high melting point (i.e., stronger bonds) will have lower lattice vibrations and specific heat value. These parameters vary non-linearly with temperature, pressure, and materials composition. Several handbooks are available that contain an extensive list of the thermophysical properties at different temperature values [52–54].

Thermal conductivity refers to the ability of material to conduct heat. In metallic materials, the addition of alloying elements decreases electron mobility and thermal conductivity. Thus, the thermal conductivity for pure metals like copper, molybdenum, or tungsten are higher than alloys like stainless steel, superalloys, or high entropy alloys. Thermal diffusivity is the ratio of thermal conductivity to density and specific heat. The product of density and specific heat provides information about the amount of heat that can be stored in a material. Thus, thermal diffusivity represents the rate of heat transfer through the material. Interestingly, copper has a lower melting point of 1356 K as compared to Inconel 718, but still needs a higher energy supply to process [55] because of its high thermal diffusivity [52].

For some materials, thermophysical properties are not readily available in handbooks, especially multicomponent alloys like high entropy alloys or graded alloy compositions. In such cases, a linear approximation method based on the weighted alloy composition can be used as a quick approximation [5]. For a binary composition, the effective density, thermal conductivity, and specific heat can be calculated as:

$$\rho_{eff} = \rho_A X_A + \rho_B X_B \tag{1}$$

$$K_{eff} = K_A X_A + K_B X_B \tag{2}$$

$$C_{p,eff} = \frac{\rho_A X_A C_{p,A} + \rho_B X_B C_{p,B}}{\rho_A X_A + \rho_B X_B} \tag{3}$$

where X_A and X_B are the elemental mole fraction, ρ_A and ρ_B are the elemental density (kg/m^3), ρ_{eff} is effective density (kg/m^3), K_A and K_B are the elemental thermal conductivity (W/mK), K_{eff} is effective thermal conductivity (W/mK), $C_{p,A}$ and $C_{p,B}$ are the elemental specific heat (J/kgK), and $C_{p,eff}$ is effective specific heat (J/kgK). Experimentally, the thermophysical properties of materials can be measured using thermal analysis. For example, melting temperature and specific heat can be measured using differential scanning calorimetry and thermal conductivity of metallic materials can be extracted using the light flash method [56,57]. Another material-dependent parameter is the laser absorption coefficient which helps in identifying the energy absorbed by the material. Unlike thermophysical properties, the value of the laser absorption coefficient is a function of laser wavelength, laser parameters, powder particle size, material composition, and powder packing density. The accurate measurement of the laser absorption coefficient is challenging, and dedicated *in situ* measurements or mathematical modelling are required [58,59].

2.2. Processing parameters

LPBF is a complex process where the solidification conditions can vary to a great extent. For example, cooling rates in the LPBF process ranges from 10^4 K/s to 10^6 K/s [2,60]. Similarly, thermal gradient (G) and solidification rate (R) can vary by two orders in magnitude. Studies have demonstrated that scan strategies and the G/R ratios can be applied to promote columnar-to-equiaxed transitions and refine the as-fabricated microstructures [61–63]. Beam shaping is another strategy that can be used to alter crystallographic texture or solidification rate in the LPBF process [64,65]. These strategies can affect the residual stress build up, microstructure formation, and eventually the mechanical properties of the material. Powder spreading, and powder contamination levels also have a direct consequence on the as-solidified microstructures and properties of the material, but similar to residual stress or microstructure features, the user has no option to control them independently. In other words, there is no dedicated parameter for controlling residual stress, or grain size in LPBF. Therefore, it becomes important to study the effects of processing parameters on laser-metal interaction and solidification conditions. An example of every independent processing parameter for the LPBF process can be found on the EOS (GmbH, Germany) website [3]. These parameters can be classified into three broad categories: (i) laser parameters, (ii) exposure settings, and (iii) machine settings as shown in Fig. 3. Within each category, the parameters can be further sub-classified either as job-specific, geometry-specific, or powder-specific. For example, coater speed influences the powder spreading and powder bed packing density, and its value depends on powder size distribution (i.e., powder-specific parameter).

In principle, all processing parameters directly or indirectly affect the part's microstructure, properties, and performance. For example, build plate temperature influences residual stresses in LPBF components, shielding gas affects the powder bed contamination and spattering, and support structure parameters dictate the quality of overhanging

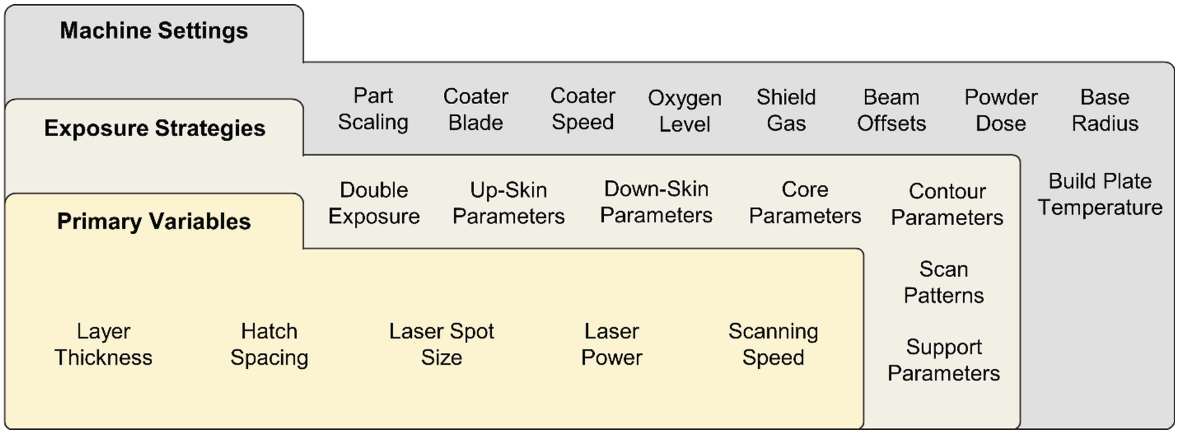

Fig. 3. The broad category of the LPBF processing parameters.

structures [9,66,67]. Most studies have focused on five primary processing variables as they dictate the amount of energy supplied to the material per unit volume and localized melting of the powder as mentioned in Fig. 2. For a moving laser beam, the laser power represents the amount of heat supply to a localized region and the scanning speed relates to the time for which the heat is supplied in a localized region. A higher value of scanning speed might reduce the print time and cost but make the part more vulnerable to defects. The cross-sectional area of the laser beam varies with height, and thus the laser spot size refers to the size of the laser beam when in focus. Some of the studies have reported a productivity increase of about 840% by using dynamic laser units to control the laser focus-defocus [68]. Laser spot size diameters for most of the AM techniques range between 50 and 600 μm. For the LPBF process, laser spot size tends to be around 50–100 μm and for the DED process, it is usually around 600 μm. It should be noted that the size of the localized melt pool is not the same as the laser spot size. The melt pool size depends upon the laser parameters, material properties, and laser absorption coefficient. Hatch spacing and layer thickness represent the distance between successive tracks and successive layers, respectively. These parameters affect the re-melting of the previously solidified materials and play an important role in obtaining good strength and interlayer bonding. To fabricate sound components, the hatch spacing, and layer thickness shouldn't exceed melt pool width and depth values, respectively. In other words, there should be sufficient melt pool overlap and penetration.

Exposure strategies enable laser parameter variations for a different part's geometric features and microstructural evolution. Down-skin and up-skin parameters refer to the laser parameters for the initial and the final few layers of the deposition, respectively. Core parameters provide information about the laser parameters for the remaining layers between up-skin and down-skin. As compared to core parameters, up-skin and down-skin layers are often deposited at a lower energy density to obtain a better surface finish [3]. Support structure enables fabrication of overhanging design features and is purposely made porous to be easily removed in post processing. The energy density requirements for support structures are about 40% that of core parameters [3]. Scan strategy sets the pattern in which the laser source moves within a layer, and hatch rotation determines the laser movement pattern between successive layers. Some of the common scan strategies are continuous line pattern, stripe pattern, island pattern, and scanning in-out pattern [66]. Each of these patterns has multiple variants with hatch angles aligned at different angles with respect to the horizontal axis. Unidirectional or bidirectional laser motion is another common variation within each scan strategy. Scan strategies affect the local heat distribution, microstructural texture, and residual stresses [7,8,69]. For example, the default scan strategy for the EOS M290 (GmbH, Germany) system is 5 mm stripe thickness with bidirectional laser motion and 67 degrees of hatch angle rotation [3]. Multiple studies have shown that patterns with shorter scan lengths, such as the island scan strategy, result in faster cooling and different residual stress states, especially around the sample edges [5,66,70].

3. Design parameters

The material properties and LPBF processing parameters combine to form a large set of input parameter required for fabricating components. Design methodologies to condense the multiple input parameters into physically significant relationships for evaluating the LPBF process have been explored and will be presented. Please note that millimeter is often used to reflect the length scale in energy density terms introduced below. In more derived analytical expressions introduced later in the text, the standard SI unit of meters is used. The difference is purely a matter of convenience.

3.1. Energy density

Multiple energy density-based parameters are proposed to provide the combined effects of laser parameters. These design parameters are helpful in visualizing and optimizing the processing conditions. Some of the commonly used energy density parameters are:

(a) *Linear energy density (LED)* gives the amount of the input energy supplied over a laser distance (in J/mm). In this case, the laser beam is considered as a point heat source. This parameter correlates laser power and scanning speed and can be expressed as:

$$Linear\ \ Energy\ \ Density\ \ (LED) = \frac{P}{V} \tag{4}$$

where P is laser power (W), and V is scanning speed (mm/s).

(b) *Areal energy density (AED)* is a modified version of the LED where the laser spot radius is incorporated to estimate the energy spread over the unit area (in J/mm^2). AED correlates all three laser parameters using the following equation:

$$Areal\ \ Energy\ \ Density\ \ (AED) = \frac{P}{Vr_l} \tag{5}$$

where P is laser power (W), V is scanning speed (mm/s), and r_l is laser spot radius (mm).

(c) *Volumetric energy density (VED)* gives the information about energy supplied per unit volume (in J/mm^3). This parameter accounts for the total energy supplied to a local region including the additional energy supplied during heating cycles. Therefore, hatch spacing and layer thickness are used in the equation to provide a better approximation of the process input parameters. VED can be calculated as:

$$Volumetric\ Energy\ Density\ (VED) = \frac{P}{VHL} \tag{6}$$

where P is laser power (W), V is scanning speed (mm/s), H is hatch spacing (mm), and L is layer thickness (mm).

(d) *Specific energy density (SED)* is a modified version of VED where instead of using hatch spacing and layer thickness values, actual melt pool dimensions were considered. To a certain degree, SED provides a better approximation of the energy supplied (in J/mm^3) since the melt pool geometry can vary locally along the build direction [4]. However, using SED as a design parameter can be challenging since melt pool geometry is difficult to measure *in situ* and it is not an independently controlled processing variable. SED can be calculated in terms of melt pool width and melt pool depth as:

$$Specific\ Energy\ Density\ (SED) = \frac{P}{Vwd} \tag{7}$$

where P is laser power (W), V is scanning speed (mm/s), w is melt pool width (mm), and d is melt pool depth (mm).

Out of all four, VED is a widely accepted design parameter by the AM community because of its simplicity. Some studies have shown a correlation between microstructure and VED [41,71]. In contrast, other studies have shown part's density variation even for the same VED [72]. Comparing the results from different studies suggests that the VED can be considered as a reliable parameter only within a certain laser power and scanning speed range. At extremely high or extremely low VED values, individual effects from laser power or scanning speed can have a dominating effect on printability.

3.2. Normalized enthalpy

Normalized enthalpy is an alternate design parameter used by the AM community. This parameter was first developed for welding processes and was later modified for the AM process [34]. Normalized enthalpy has a few merits over energy density. It combines both material properties and laser parameters which are essential for describing the laser-metal interaction. In a recent study, normalized enthalpy combined with FEM models were used to translate the optimal processing parameter from one alloy to another [73]. Another study has shown a dependence between the normalized enthalpy and melt pool depth [34]. Thus, the conduction to keyhole mode heat transfer can be estimated using this parameter. Normalized enthalpy can be calculated using the following equation:

$$\frac{\Delta H}{H_s} = \frac{\alpha P}{\pi H_s \sqrt{D_t V r_l^3}} = \frac{\alpha P}{\pi \rho C_p T_m \sqrt{D_t V r_l^3}} \tag{8}$$

where P is laser power (W), V is scanning speed (m/s), r_l is laser spot radius (m), T_m is melting temperature (K), ρ is density (kg/m^3), C_p is specific heat (J/kgK), D_t is thermal diffusivity (m^2/s), and α is laser absorptivity. However, unlike energy density, normalized enthalpy does not account for hatch spacing or layer thickness. In other words, this parameter will not provide any information about defects such as lack of fusion, which are heavily dependent upon the melt pool layers overlapping. Also, the normalized enthalpy depends on the laser absorption coefficient which can be challenging to measure experimentally.

3.3. Dimensionless numbers

Dimensionless numbers are derived from dimensional analyses using the Buckingham Pi theorem [74]. Some of the popular dimensionless numbers relevant to AM processes are the Fourier number, the Peclet number, the Marangoni number, the Stefan number, the Weber number, and the non-dimensional heat input, etc. [75]. The Fourier number provides information about the rate of heat dissipated with respect to the rate of heat stored in the part. The Fourier number can be expressed as:

$$Fo = \frac{D_t \tau}{l^2} = \frac{D_t}{Vl} \tag{9}$$

where, D_t is thermal diffusivity (m^2/s), τ is characteristic time (s), l is melt pool length (m), and V is scanning speed (m/s). A higher value of the Fourier number is desirable to reduce heat storage, thermal strain, and residual stress. The Peclet number is defined as:

$$Pe = \frac{U_{max} L_c}{D_t} \tag{10}$$

where, D_t is thermal diffusivity (m^2/s), U_{max} is the maximum velocity of molten metal inside the fusion zone (m/s), and L_c is the characteristic length such as length or width of the melt pool (m). At a lower Peclet number, the conduction mode of heat transfer is dominant, while at a higher Peclet number, the convective mode of heat transfer starts to dominate. Similarly, other dimensionless numbers provide information such as melting efficiency, capillary effects, and extent of the Marangoni effects within the melt pool [18,75]. However, most of these dimensionless numbers cannot be used as a sole design parameter to identify the processing window of the LPBF process. In a recent study by Rankouhi *et al.* [45], a dimensionless energy density was used for determining the processing window for the LPBF process. The dimensionless energy density can be expressed in terms of materials properties and laser parameters as:

$$E_{dim} = \frac{C_p P}{K V^2 H} \tag{11}$$

where P is laser power (W), V is scanning speed (m/s), H is hatch spacing (m), K is thermal conductivity (W/mK) and C_p is specific heat (J/kgK). There are two major benefits of using dimensionless energy density as a design parameter. First, it can be applied to translate the optimal processing parameters across different materials or LPBF systems. Second, all the variables needed to calculate the dimensionless energy density are either material's constant or independently controlled process variables.

It is also interesting to compare the three design parameters from Eqs. (6), (8), and (11) with respect to P and V. Each design parameter can be expressed as a function of P and $1/V^n$ i.e.:

$$Design\ parameters \propto \frac{P}{V^n} \tag{12}$$

For $n = 1$, the design parameter is volumetric energy density (linear), for $n = 0.5$ it is normalized enthalpy equation (squared root), and for $n = 2$ it is dimensionless energy density (squared). A schematic diagram of this inter-dependency is shown in Fig. 4. However, the LPBF process is complex, and the melt pool dynamics may or may not be fully captured in any of these design parameters depending upon the range of parameters being studied. It is also possible that different design parameters would be appropriate for different ranges of power-velocity combinations. Nevertheless, design parameters are useful for quick approximation and reducing the dimensionality of the LPBF processing variables.

4. Analytical melt pool models

Upon exposure to the laser energy, the powder's surface temperature rises rapidly. In a localized region, the temperature exceeds the melting temperature forming a melt pool boundary. The supplied heat also raises the temperature in the vicinity of the melt pool. These regions are known as heat-affected zones. The melt pool shape and size have a direct

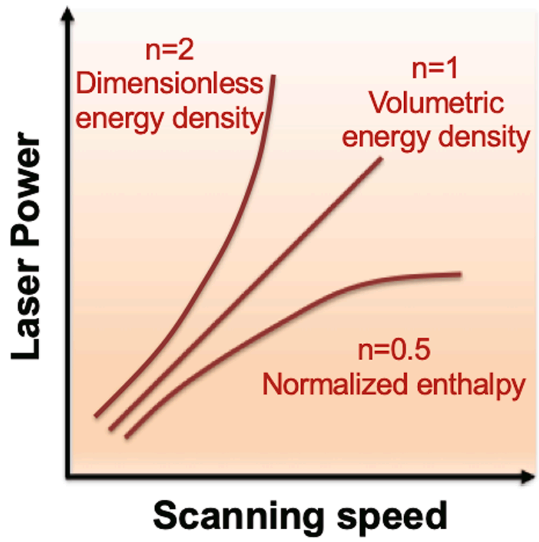

Fig. 4. Comparison of the three commonly used design parameters from Eq. (6), (8), and (11), showcasing the difference in the inter-dependency between laser power and scanning speed.

consequence on porosity defects, solidification conditions, and microstructural features. The two governing forces that control the melt pool flow are the Marangoni force and recoil pressure [33,76]. Other forces such as buoyancy force, gravitational force, and Lorentz force have negligible effects on the melt pool flow. The Marangoni force is a thermo-capillary effect that occurs due to large thermal gradients and temperature dependent surface tension [77,78]. It pushes the molten metal to flow from high temperature to low temperature regions, making the melt pool flow vigorously within the fusion boundary. Recoil pressure originates from the rapid evaporation of alloying elements from the melt pool surface. This force pushes the liquid metal inwards creating a vapor cavity known as vapor depression zone [76]. At a high energy density, the peak temperature, Marangoni force, and recoil pressure increase, leading to an unstable melt pool [21]. Thus, the melt pool geometry can become shallow, deep, or semi-spherical depending on the heat supply and processing variables [27,79].

Techniques such as *in situ* synchrotron x-ray imaging, infrared thermal imaging, FEM simulations, and optical microscopy have been used to study and measure the melt pool geometry and heat distribution as shown in Fig. 5. Analytical models use the thermophysical data and processing parameters to estimate the melt pool geometry at given processing conditions. Unlike *in situ* imaging and FEM simulations, these models provide time efficient, computationally inexpensive, and quick approximation of the melt pool geometry. Details of these analytical models based on the Rosenthal equations and scaling laws are discussed below.

4.1. Rosenthal equations

Heat distribution in a solid with a moving heat source (e.g. welding processes) can be calculated using the Rosenthal equations [82]. In the study by Tang *et al.* [10] the Rosenthal equations were used to obtain an analytical expression of melt pool geometry i.e., the region for which the local temperature is equal to the alloy's melting temperature. Some of the key assumptions made to derive the analytical equations are: (i) the heat flow is two-dimensional, (ii) there are negligible effects from convective heat transfer i.e., only conductive mode of heat transfer is considered, (iii) the laser beam acts as a point source, (iv) specific heat and thermal conductivity are temperature independent, and (v) the melt pool is semi-elliptical in shape. The width of the melt pool can be expressed as:

$$Width(w) = \sqrt{\frac{8\alpha P}{\pi e \rho C_p V (T_m - T_o)}} \tag{13}$$

Since a semi-elliptical melt pool shape was assumed, the melt pool depth can be expressed as:

$$Depth(d) = \frac{Width(w)}{2} = \sqrt{\frac{2\alpha P}{\pi e \rho C_p V (T_m - T_o)}} \tag{14}$$

where P is laser power (W), V is scanning speed (m/s), T_m is melting temperature (K), T_o is the far-away temperature (K), assumed to be the

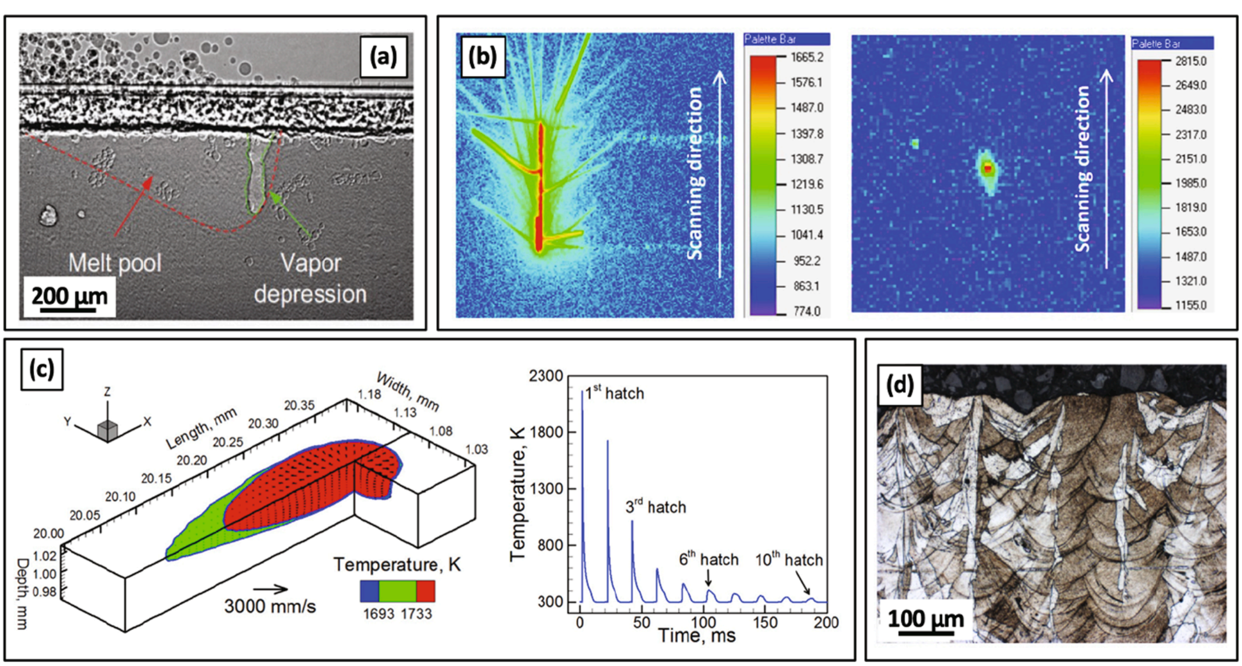

Fig. 5. Different approaches are used to study melt pool (a) *in situ* synchrotron x-ray imaging [23] (b) *in situ* infrared thermal imaging at two different temperature ranges [80] (c) FEM-based simulations [31] (d) optical microscopy of as-fabricated materials [81].

build plate temperature, ρ is density (kg/m^3), C_p is specific heat (J/kgK), and α is laser absorptivity.

4.2. Scaling laws

In the study by Rubenchik *et al.* [36], a scaling law analysis based on the Egar-Tsai (ET) thermal model was performed to calculate the melt pool geometry. Some of the key model assumptions are: (i) melt flow and thermo-capillary effects are neglected, (ii) the effects of the heat of fusion on temperature distribution is not considered, and (iii) specific heat and thermal conductivity are temperature independent. From this model the melting process can be characterized in terms of two dimensionless numbers p and B. The expression for p and B is given as:

$$p = \frac{D_t}{Vr_l} \tag{15}$$

$$B = \frac{\alpha P}{\pi\rho C_p T_m \sqrt{D_t V {r_l}^3}} \tag{16}$$

where P is laser power (W), V is scanning speed (m/s), r_l is laser spot radius (m), T_m is melting temperature (K), ρ is density (kg/m^3), C_p is specific heat (J/kgK), D_t is thermal diffusivity (m^2/s), and α is laser absorptivity. The dimensionless number p represents the overall shape of the melt pool, and it is the ratio of the thermal diffusion length to the dwell time. Therefore, a lower value of p means the melt pool is shallow and long. The dimensionless number B represents the ratio of the input laser energy to the enthalpy. For a well-developed melt pool, both p and B have a value greater than 1. The melt pool dimensions including width, depth, and length are defined in terms of dimensionless numbers p and B and can be expressed as:

$$Depth(d) = \frac{r_l}{\sqrt{p}}\left[\begin{array}{c} 0.008 - 0.0048B - 0.047p - 0.099Bp \\ +plnp(0.32 + 0.015B) + lnB(0.0056 - 0.89p + 0.29plnp) \end{array}\right] \tag{17}$$

$$Length(l) = \frac{r_l}{p^2}\left[\begin{array}{c} 0.0053 - 0.21pB + 1.3p^2 - (0.11 + 0.17B)p^2 lnp \\ +B(-0.0062 + 0.23p + 0.75p^2) \end{array}\right] \tag{18}$$

$$Width(w) = \frac{r_l}{Bp^3}\left[\begin{array}{c} 0.0021 - 0.047p + 0.34p^2 - 1.9p^3 - 0.33p^4 + \\ B(0.00066 - 0.007p - 0.00059p^2 + 2.8p^3 - 0.12p^4) + \\ B^2(-0.0007 + 0.015p - 0.12p^2 + 0.59p^3 - 0.023p^4) \\ +B^3(0.00001 - 0.00022p + 0.002p^2 - 0.0085p^3 + 0.0014p^4) \end{array}\right] \tag{19}$$

In another analysis, Fabbro *et al.* [83] utilized scaling laws to predict the melt pool aspect ratio (depth-to-width ratio). The following simplifying assumptions were made in constructing this model: (i) the melt pool is cylindrical in shape with diameter and length equal to the laser spot size diameter and melt pool depth, respectively, (ii) the temperature of the keyhole wall is constant and equal to the evaporation temperature of the material, and (iii) the melt pool is moving across a substrate with a constant temperature and at a constant speed. The melt pool aspect ratio is defined as:

$$R = \frac{d}{w} = \frac{R_0}{1 + (V/V_0)} \tag{20}$$

where,

$$R_0 = \frac{\alpha P}{n d_l K (T_b - T_o)} \tag{21}$$

$$V_0 = \frac{2nK}{m d_l \rho C_p} \tag{22}$$

Here, R is the melt pool aspect ratio (depth-to-width), d is melt pool depth (m), w is melt pool width (m), V is laser scanning speed (m/s), α is laser absorptivity, P is laser power (W), d_l is laser spot diameter (m), K is thermal conductivity (W/mK), T_b is evaporation temperature (K), T_o is substrate temperature (K), ρ is density (kg/m^3), C_p is specific heat (J/kgK), m and n are linear coefficients that depend on the Pe number (refer to Eq. (10)) and can be approximated analytically (to determine the values of m and n, readers are referred to [83]).

The melt pool aspect ratio can be used to determine the transition from conduction to keyhole mode of heat transfer in the LPBF process. It can be assumed that melt pool shifts from conduction to keyhole for $R \geq 1$. This transition plays a vital role in LPBF processing of highly reflective materials such as copper. Jadhav *et al.* [84] have used the melt pool aspect ratio to determine the laser power and scan speed values where the keyhole mode of heat transfer can be achieved in LPBF processing of pure copper.

5. Criteria for porosity threshold

Several types of defects can be observed in the as-fabricated AM components such as porosities, contamination, cracking, delamination, distortion, surface roughness, microstructure inhomogeneity, and crystallographic anisotropy. This section specifically focuses on porosity and different analytical models to determine the porosity thresholds for lack of fusion, balling, keyholing, and partially melted powder particles. These analytical models are based upon the melt pool geometry and are either derived from the mathematical considerations or the experimental observations. Determination of these threshold values are essential to identify the processing window for any alloy.

5.1. Lack-of-fusion offset

Lack of fusion is one of the most studied porosity defects. The cross-sectional surface of the LPBF-fabricated 316L stainless steel containing a lack-of-fusion defect is shown in Fig. 6a. These defects are detrimental to the properties because of their high aspect ratio and sharp edges. Their volume fraction increases with a decrease in energy supply [15,17]. But even in the case of a fully developed melt pool, a substantial fraction of lack-of-fusion defect can exist due to insufficient melt pool overlapping and high hatch spacing. In the study by Tang *et al.* [85] considering a dual semi-elliptical melt pool, a geometric model was proposed for the lack of fusion criterion as shown below:

$$\left(\frac{H}{w}\right)^2 + \left(\frac{L}{d}\right)^2 \leq 1 \tag{23}$$

where H is the hatch spacing (mm), L is the layer thickness (mm), w is the melt pool width (mm), and d is the melt pool depth (mm). According to this model, the ratio of the melt pool width to hatch spacing and melt pool depth to layer height should be high enough to provide sufficient overlapping. In another recent study [17], an expression for lack-of-fusion (LOF) index was applied to estimate the porosity defect fraction. It can be calculated in terms of laser parameters, melt pool geometry, and materials properties as:

$$\text{LOF index} = \rho(C_p\Delta T + \Delta H) \times \frac{\pi {r_l}^2 V}{\alpha P} \times Fo \times \frac{L}{d} \times \left(\frac{H}{w}\right)^2 \tag{24}$$

where P is laser power (W), V is scanning speed (m/s), H is hatch spacing (m), L is layer thickness (m), ΔT is the difference between peak temperature and melting temperature (K), ΔH is the latent heat of the material (J/kg), ρ is density (kg/m^3), C_p is specific heat (J/kgK), α is laser absorptivity, d is the melt pool depth (m), w is the melt pool width (m), and Fo is the Fourier number (dimensionless). As a rule of thumb, the penetration depth of the melt pool should be 50% higher than the layer thickness (i.e., $d/L > 1.5$) to avoid lack-of-fusion defects and improve interlayer bonding [35]. Also, the fraction of the lack-of-fusion defects increases linearly with the LOF index and can be calculated using:

$$Volume\ of\ the\ LOF\ defects = 15.3 \times LOFindex \tag{25}$$

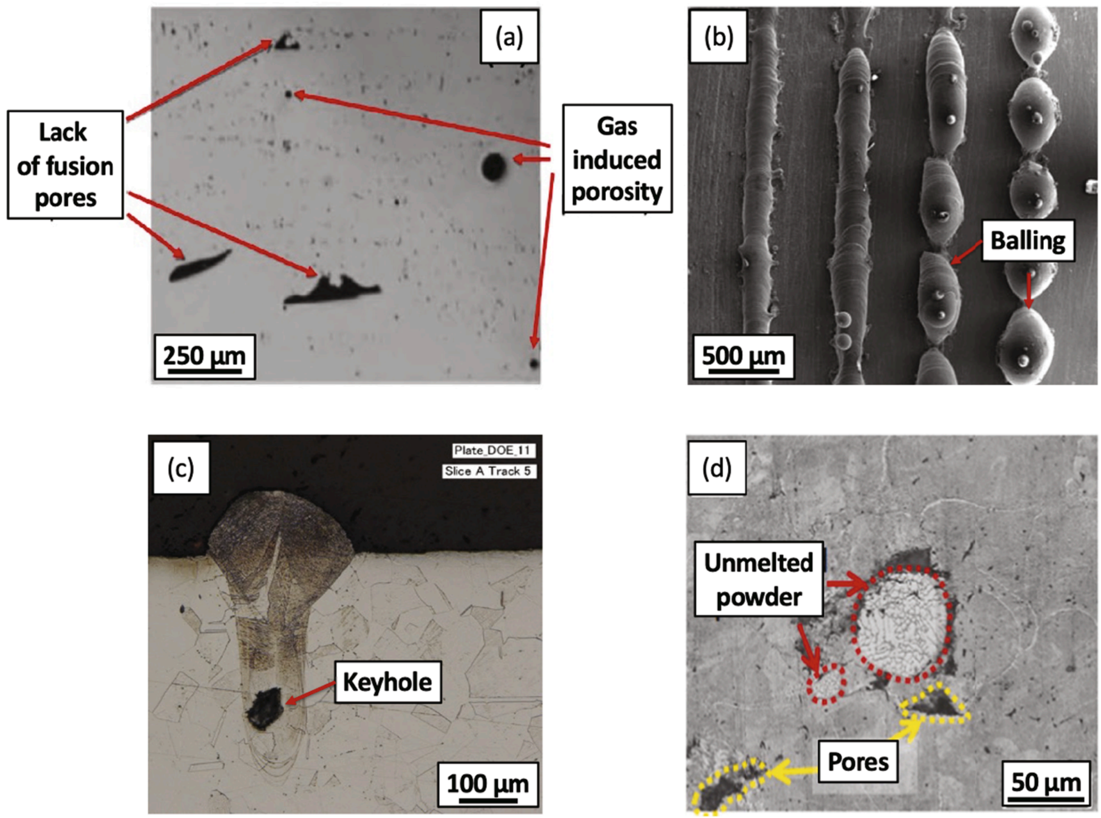

Fig. 6. Cross sectional images of parts containing (a) lack-of-fusion defects [86] (b) balling [19] (c) keyhole porosities [34] and (d) unmelted powder particles [48].

5.2. Balling threshold

Balling defects are a result of dynamic instability where the humping or irregular beads form along the melt pool tracks. The as-deposited surface of the balling defects is shown in Fig. 6b. These defects are influenced by the fluid dynamics, surface tension, and morphology of the melt pool [4]. Balling also occurs in the processes involving fluid flows such as welding [87,88]. When the Plateau-Rayleigh instability criterion is satisfied, that is the ratio of the melt pool length and width exceeds the critical value of pi, the molten metal tries to lower its surface energy by forming spherical droplets instead of spreading over the base plate [5,20]. In the work by Yadroitsev *et al.* [89] a modified version of the balling criterion was proposed for single track formation in the LPBF process. This model considered the added stability provided by the horizontal segmental melt pool tracks and can be represented by:

$$\frac{Length(l)}{Width(w)} < \sqrt{\frac{3}{2}}\pi \ \ or \ \ 3.8 \tag{26}$$

Similarly, different critical values of the dynamic instabilities have been reported ranging from 2.3 to 6.24 depending upon the application. At high laser power and scanning speed, the melt pool has a lower width and an elongated tail end, and such processing conditions promote the balling defect [5,88].

5.3. Keyholing threshold

Keyhole porosities are spherical types of defects which occur due to melt pool instability at excessive heat supply. These pores often form in pairs and can be observed much below the powder surface [5,21]. A cross-sectional image of a specimen containing keyhole porosities is shown in Fig. 6c. An increase in laser power results in a gradual increase of the melt pool depth, and beyond a transition point, the depth starts to increase rapidly. This transition point is when heat transfer changes from conduction to keyhole mode. [34,90]. At larger melt pool depths, the chances of gas entrapment from the vapor depression zone increases. Thus, the frequency of keyhole porosities also increases with energy density. Studies performed on *in situ* synchrotron x-ray techniques show that a higher melt pool front angle leads to keyhole porosities [21]. In the study by King *et al.* [34] an experimental correlation between the normalized enthalpy and keyhole threshold was reported. The keyhole threshold can be approximated by taking the ratios of the melting temperature (T_m) and boiling temperature (T_b) as shown below:

$$\frac{\Delta H}{H_s} = \frac{\alpha P}{\pi \rho C_p T_m \sqrt{D_t V r_l^3}} < \frac{\pi T_b}{T_m} \tag{27}$$

where P is laser power (W), V is scanning speed (m/s), r_l is laser spot radius (m), T_m is melting temperature (K), T_b is boiling temperature (K), ρ is density (kg/m^3), C_p is specific heat (J/kgK), D_t is thermal diffusivity (m^2/s), and α is laser absorptivity.

Data from different laser beam diameters suggest that the keyhole threshold for various alloys ranges between a normalized enthalpy value of 6 to 30. There are also some studies that predict the keyhole threshold based on the melt pool geometry [14]. Experimental data suggest that keyhole porosities can be avoided by increasing the width to the depth ratio as given below:

$$\frac{Width(w)}{Depth(d)} > 1.5 \tag{28}$$

With an increase in energy supply, the melt pool geometry changes from a shallow semi-spherical shape to a deep cylindrical shape. Therefore, Eq. (28) suggests that keyholing occurs at higher energy densities.

5.4. Under-melted or partially melted threshold

Another potential processing defect can be the presence of under-melted or partially melted particles in the part. Under the scenarios of a low energy supply or a material with a low laser absorption coefficient, the fraction of under-melted particles can be high [48]. The cross-sectional surface of a component containing under-melted particles is shown in Fig. 6d. The issue of partially melted particles become more prevalent when two or more elemental powders having very different melting temperatures are mixed to get a desired alloy composition. The surface temperature can be estimated in terms of normalized enthalpy as given below:

$$T_{surface} = \frac{T_m \Delta H}{H_s} = \frac{\alpha P}{\pi \rho C_p \sqrt{D_t V r_l^3}} \tag{29}$$

where P is laser power (W), V is scanning speed (m/s), r_l is laser spot radius (m), T_m is melting temperature (K), ρ is density (kg/m^3), C_p is specific heat (J/kgK), D_t is thermal diffusivity (m^2/s), and α is laser absorptivity.

The under-melted and partially melted particles will get infused in a component when the surface temperature is lower than the melting temperature of the alloy i.e.:

$$T_{surface} = \frac{\alpha P}{\pi \rho C_p \sqrt{D_t V r_l^3}} < T_m \tag{30}$$

6. Additional analytical models

There are other analytical models and dimensionless numbers that do not necessarily contribute to construction of the processing maps but are relevant to the LPBF process. In this section, models to estimate vapor pressure, compositional change, thermal strain, depression depth, solidification cracking, and dendritic arm spacing are discussed briefly.

6.1. Vapor pressure and composition change

In the case of excessive heat supply, the peak temperature can easily exceed the material's boiling temperature leading to high vapor pressure and material volatilization. The saturated vapor pressure (P_s) can be calculated using the expression [34]:

$$P_s = P^* exp\left[\lambda\left(\frac{1}{T_b} - \frac{1}{T_s}\right)\right] \tag{31}$$

where P^* is the atmospheric or the chamber pressure (Pa), λ is the evaporation energy per atom (eV), T_b is the boiling temperature (K), and T_s is the surface temperature (K) (refer to Eq. (29)). Similarly, the recoil pressure (P_r) can be estimated to be around 0.56 times the saturated vapor pressure (P_s) [34]. In some alloys, especially when the boiling temperature of two alloying elements is significantly different, high vapor pressure can lead to variation in the compositional concentration. The vaporization flux (J_i) can be estimated using the Langmuir equations [35]:

$$J_i = \frac{P_i}{\sqrt{2\pi M_i R T_s}} \tag{32}$$

where P_i is vapor pressure for alloying element i (Pa), M_i is the molecular weight of element i (kg/mol), R is the molar gas constant (J/molK), and T_s is the surface temperature (K) (refer to Eq. (29)). Finally, the amount of material vaporized (Δm_i) can be calculated using the expression [35]:

$$\Delta m_i = \frac{l A_s J_i}{V} \tag{33}$$

where V is the scanning speed (m/s), A_s is the melt pool cross-sectional area (m^2), l is the track length (m), and J_i is the vaporization flux (mole/m^{2}s). Calculating the actual value of the melt pool area can be complicated. Nevertheless, Eqs. (17)–(19) can be applied to get rough estimates of the melt pool geometry. The above equations can assist in defining the optimal processing parameters by avoiding material volatilization and fabricating parts with desired chemical composition.

6.2. Thermal strain

Residual stress in AM components originates from the spatial variation in thermal gradient and nonuniform thermal expansion and contraction [9,91]. For example, during solidification of the liquid metal, the new layer solidifies quickly and contracts at a faster rate than the layers beneath it. The geometrical constraints impose tensile stresses in the top layer and compressive stresses in the subsequent bottom layers [9]. These macroscopic stresses (long gradients of plastic deformation) can cause cracking, warpage, and distortion of AM components. The extent of the part's distortion can be estimated in terms of thermal strain (ϵ_t) and can be expressed in terms of materials property and processing parameters as [35]:

$$\epsilon_t = \frac{0.9\beta}{EIF_o} \times \frac{\Delta T \tau}{\sqrt{\rho}} \times LED^{3/2} \tag{34}$$

where LED is linear energy density (J/m) (refer to Eq. (4)), ρ is density (Kg/m^3), ΔT is the temperature difference between the peak temperature and the solidus temperature (K), τ is the characteristic time (s), β is the coefficient of linear expansion, E is elastic modulus (MPa), I is the moment of inertia of the base plate (Kgm^2), and Fo is the dimensionless Fourier number. It is evident from the above equation that the thermal strain increases with an increase in LED and a control over the laser parameters, thermal gradient, and cooling rates can reduce the residual stress within AM components. Another factor which can influence residual stress is part geometry, more specifically changes in moment of inertia [4,32,92]. For example, warpage of a rectangular bar occurs along it's long axis and the propensity of warpage increases with rectangular bar's aspect ratio.

In polycrystalline materials, another subcategory of residual stress (i.e., microscopic residual stress) prevails due to microstructural heterogeneity [93]. Local lattice mismatch, impurities, and high dislocation densities lead to elastic stress fields and differences in slip behavior. Studies have highlighted that dislocation density increases with an increase in energy density [94,95]. However, individual effect of macroscopic and microscopic residual stress cannot be estimated using Eq. (34). Process parameters including hatch spacing and layer thickness relate to the number of heating and cooling cycles. Each cycle causes annealing effects in the surrounding regions resulting in annihilation of dislocations. It is important to note that the above analytical expression does not account for annealing effects and stress relaxation. To completely estimate residual stresses at different processing conditions more rigorous experimental and computational techniques should be applied [9,91,96,97].

6.3. Depression depth

A vapor cavity (i.e., vapor depression depth) forms due to the evaporation from the melt pool surface and a high recoil pressure exerting inward forces on the melt pool. A very large depression depth increases the probability of keyhole porosity formation. Several *in situ* synchrotron x-ray studies have been performed to study the depression depth and laser-metal interaction. In the study by Gan *et al.* [98] low dimensional scaling laws have been presented to estimate the depression depth and porosity fraction. The depression depth was defined in terms of a dimensionless keyhole number (Ke) which can be expressed as:

$$Ke = \frac{\alpha(P - P_o)}{V d_l^2}\left(\frac{1}{\rho C_p (T_b - T_o)}\right)\left(\frac{T_m - T_o}{T_b - T_o}\right) \tag{35}$$

where P is laser power (W), V is scanning speed (m/s), d_l is laser spot diameter (m), P_o is the minimum laser power able to generate a keyhole (W), α is laser absorption coefficient, ρ is density (Kg/m^3), C_p is specific heat (J/kgK), T_m is melting temperature (K), T_b is boiling temperature (K), and T_o is the build plate temperature (K). All the experimental data obtained from the *in situ* studies for different alloys correlated well with the depression depth (e^*) and Keyhole number. This correlation is expressed as:

$$e^* = 5.12 - 5.07exp(-Ke) \tag{36}$$

6.4. Solidification cracking

The ability of a material to resist solidification cracking (or hot cracking) depends upon its thermodynamic properties and the solidification conditions. One of the rough parameters that can be used to quickly estimate the material's susceptibility to solidification cracking is the freezing temperature range i.e., the difference between the liquidus and the solidus temperature. A material with a high freezing temperature range takes longer to solidify with subsequent formation of thin liquid films between the grain boundaries making them prone to solidification cracking. Other metallurgical factors such as the presence of low-melting eutectics, solidification morphology, and surface tension of the grain boundary liquid also influence the material's tolerance to solidification cracking. The crude estimates from the freezing temperature range criterion are not effective to design alloy compositions, especially for welding and AM processes because of the high cooling rates and metastable solidification. In studies by Kou [99,100], an index was presented to calculate the alloy's susceptibility to solidification cracking. According to this criterion, the cracking depends upon the maximum steepness of the curve between temperature (T) and square root of the fraction solid $(f_s)^{1/2}$ and it is expressed as:

$$Kou\ \ index = \frac{dT}{d(f_s)^{1/2}} near(f_s)^{1/2} = 1 \tag{37}$$

where T is temperature (K) and f_s is solid fraction of semi-solid region. The lower the steepness, the higher the material's resistance to solidification cracking would be. This parameter can be calculated using any commercial thermodynamic software package provided that the thermodynamic database is available. In the study by Tang et al [101], a modified version of Kou index was presented by combining the effects from the solidification kinetics, solidification shrinkage, and high temperature materials property. The modified index can be expressed as:

$$Modified\ \ index1 = \frac{dT}{d(f_s)^{1/2}} \frac{1}{\sqrt{1-s_f}} \tag{38}$$

$$Modified\ \ index2 = \frac{dT}{d(f_s)^{1/2}} \frac{1}{\bar{E}} \tag{39}$$

where T is temperature (K), f_s is solid fraction of semi-solid region, s_f is shrinkage factor, and $\bar{E}$ is materials toughness (MPa) near the solidus temperature. Shrinkage factor represents the ratio of the materials density at liquidous to materials density at solidus temperature. The material with high solidification shrinkage will have more susceptibility towards solidification cracking. Other index is a function of material toughness i.e., yield strength multiplied by fracture strain. A material possessing high material toughness at elevated temperatures will show better resistance to solidification cracking. In the comparative analysis between the above index, modified index 2 (Eq. (39)) provided the better correlations with the experimental and synchrotron results.

6.5. Cooling rate and dendritic arm spacing

The average cooling rate can be estimated based on heat distribution calculations and Rosenthal equations. In the study by Tang *et al.* [102], the average cooling rate (ε) was expressed as:

$$\varepsilon = 2\pi K(T_{solidus} - T_o)(T_{liquidus} - T_o)\frac{V}{P} \tag{40}$$

Where P is laser power (W), V is scanning speed (m/s), K is thermal conductivity (W/mK), T_o is build plate temperature (K), and $T_{solidus}$ and $T_{liquidus}$ are the solidus and liquidus temperature, respectively (K). It is evident from the above equation that the cooling rate depends upon laser power and scanning speed. The average cooling rate increases with a decrease in linear energy density (refer to Eq. (4)). Experimental observations also support the inverse dependence of energy density and cooling rate [4,103]. Also, the above equation implies that the material with higher thermal conductivity will exhibit a greater cooling rate dependence with the laser parameters.

The dendritic arm spacing is directly related to the average cooling rate [104–106]. Generally, the secondary dendritic arm spacing (λ_{SDAS}) can be calculated using the cooling rate value as:

$$\lambda_{SDAS} = a\varepsilon^{-b} \tag{41}$$

Where ε is cooling rate (K/s), and a and b are material constants. For example, the values of a and b for 316L stainless steel are 25 and -0.28 respectively [107,108]. An estimate of the primary dendritic arm spacing (λ_{PDAS}) can also be made using the cooling rate, but these values show marginal deviation as the PDAS continues to coarsen during the solidification process. In as-solidified components, the yield strength and hardness are correlated with the dendritic arm spacing values using a Hall-Petch equation as:

$$\sigma_y = \sigma_o + K(\lambda_{SDAS})^{-0.5} \tag{42}$$

where σ_y is yield stress (MPa), σ_o is internal frictional stress (lattice resistance to dislocation motion) (MPa), K is strengthening coefficient, and λ_{SDAS} is secondary dendritic arm spacing (μm). For most alloys, the hardness can be approximated by taking one-third of the yield strength [109]. Thus, the as-solidified microstructures and properties are related to the laser-metal interaction (Eqs. 40–42). Therefore, a better control over the process variables can help to refine the microstructures and tailor the material's properties.

7. Predictive processing maps

Section 5 introduced four types of porosity defects and their respective analytical models and threshold criteria. These thresholds can be combined with analytical expressions reviewed in this work to obtain comprehensive processing maps for different materials. The purpose of these maps is to provide an overview of the processing space that can be constructed analytically and without the need to conduct experiments to obtain data. The selected equations to construct these maps use *a priori* parameters as opposed to parameters that require *in situ* measurements. Further refinement maybe possible with the *in situ* experimental data gathered from other expressions introduced in this work. Microstructural features such as hot short cracking, texture, and grain size will most likely be a subset within these processing bounds. Tailoring of these features will only be possible once the primary processing conditions and their associated defects are identified.

7.1. PV map construction

With consideration of the analytical expressions presented in this review, the predictive PV processing maps were constructed for eight different elemental metals and alloys i.e., 316L stainless steel, Cantor alloy, Inconel 718, Haynes 282, Copper, Tantalum, Molybdenum, and Tungsten. The thermophysical properties required to generate these PV maps are shown in Table 1. This group of materials are commercially

Table 1
Thermophysical properties of commercial metals and alloys [52].

Material	Laser absorption coeff.	Melting temp (K)	Density (kg/m^3)	Specific heat (J/kg K)	Thermal conductivity (W/mk)	Thermal diffusivity (mm^2/s)
316L SS	0.5	1683	7895	450	13.8	3.90
Cantor	0.4	1349	8042	430	12.0	3.47
IN 718	0.5	1609	8150	435	11.4	3.20
Haynes 282	0.6	1648	8300	436	10.3	2.88
Cu	0.3	1356	8930	384	399.0	115.00
Ta	0.4	3269	16,650	140	57.5	25.00
Mo	0.4	2893	10,220	250	142.0	56.00
W	0.5	3693	19,250	130	164.0	66.00

available for the LPBF process, and they represent a broad range of thermophysical properties. For example, tungsten has a high melting temperature and copper has a high thermal conductivity. In addition, some of these materials are single-phase alloys while others are precipitation hardened alloys. Thermal diffusivity for these materials ranges from 2.88 to 115 mm^2/s which differs by two orders in magnitude. Values reported in Table 1 are at room temperature and assumed to be temperature independent. In addition, changes in the physical properties due to differences in powder processing is neglected. For example, it might be possible that the feedstock powder particles may have a lower density (due to gas porosities or metastable phases) than their wrought counterparts. Finally, an approximation of the laser absorption coefficient was made by taking the average values of the reported ranges. These values are difficult to precisely measure and are dependent on the laser parameters and powder size distribution [58].

PV maps provide a visual snapshot of the LPBF processing regime. Processing maps consist of five distinct regions: keyhole (blue), balling (pink), partially melted powders (grey), lack of fusion (red), and optimal (green) as shown in Fig. 7. These PV maps were plotted at constant hatch spacing, layer thickness, and laser spot size (infrared fiber laser with ~ 100 μm spot size) for each material in accordance with some of the available experimental data in the literature. A comparison between the two is presented in Section 7.2. Additionally, constant VED lines are plotted to highlight the overall input energy required to process these materials at each region. The VED values vary from 25 to 600 J/mm^3.

The lack-of-fusion region was calculated using the Rosenthal-based melt pool equations (Eqs. (13) and (14)) and the LOF offset criterion (Eq. (23)). Rosenthal equations were derived assuming semi-elliptical melt pool geometry (i.e., the melt pool depth is twice the melt pool width). These assumptions are valid only at lower energy density values and thus they are appropriate to calculate the LOF regions. The balling region was calculated using the melt pool estimates from the Egar-Tsai models (Eqs. (17), (18), and (19)) combined with the balling criterion (Eq. (26)). Similarly, the calculations for the keyhole threshold (Eq. (27)) and under-melted regions (Eq. (30)) were based on the normalized enthalpy values (Eq. (8)). Finally, the optimal region was estimated from the dimensionless energy density (Eq. (11)) where the dimensionless energy density values of 61 and 146 represent the lower and upper processing boundaries respectively. Within these ranges, a high relative density of 99.5% was observed for different metals and alloys [45]. The selected equations and their corresponding thresholds are summarized in Table 2.

In certain sections of PV maps, overlapping of processing regimes can occur. For example, lack-of-fusion and under-melted regions overlap at lower energy density regions. In principle, all the processing conditions that result in under-melted particles will also show a lack-of-fusion porosity. So, the under-melted region is expected to be within the lack-of-fusion region. However, due to different analytical models used for each defect, the under-melted region extends outside the lack-of-fusion region. The deviation in the prediction is due to: (i) inability to precisely predict the melt pool geometry at different processing conditions, and (ii) uncertainty in the threshold values for different defect criteria. Similarly, the overlapping of keyhole and balling regions occurs at high power and high velocity values. The overlap of different regions also suggests that individual criteria is not sufficient to completely bound the formation of porosity defects.

It is important to note that the mapped regions are a result of simplified analytical models. As a result, they should be interpreted as coarse predictions rather than absolutes. Any processing condition that does not lie within the lack-of-fusion (red), under-melted (black), balling (pink), and keyholing (blue) regions is less likely to show any porosity defects during fabrication. In other words, the probability of defect formation increases as the corresponding defect boundary is crossed. Similarly, the optimal (green) region represents the set of processing parameters that are least likely to cause porosity defects.

Processing maps for 316L stainless steel and Cantor alloy are shown in Fig. 7a and b. These maps are plotted at a hatch spacing of 0.11 and 0.08 mm and a layer thickness of 0.02 and 0.05 mm. Both alloys are based on the Fe-Cr-Mn elemental constituent with 316L stainless steel having a Fe-rich composition and the Cantor alloy having an equiatomic composition. The middle of the optimal processing window for 316L and Cantor alloy is around 200 W and 800 mm/s and 250 W and 850 mm/s, respectively.

Processing maps for Inconel 718 and Haynes 282 alloys are shown in Fig. 7c and d. These maps are plotted at a hatch spacing of 0.11 mm and a layer thickness of 0.04 mm. Both alloys are precipitation hardened Ni-base superalloys and have similar alloy compositions. The lack-of-fusion region estimated for Haynes 282 is much bigger as compared to Inconel 718. However, the middle of the processing window for both alloys is around 250 W and 950 mm/s. In comparison to the 316L stainless steel, the Ni-base superalloys require about 50% less input energy density, primarily due to the use of a thicker layer thickness in the calculations.

Processing maps for copper and refractory metals are shown in Fig. 7e-h. Interestingly, the keyhole and balling region is not expected in any of these materials within the calculated processing ranges as shown in Fig. 7e-h. All these metals require very high energy density to fabricate, and the processing window is skewed towards the high power of 360–400 W and low scanning speed of 200–250 mm/s. Refractory metals have high melting temperatures, therefore, higher input energy is needed to melt them. Although the melting temperature of copper is low, the metal has a high thermal diffusivity and low laser absorption coefficient in the infrared range. Most of the heat absorbed by copper powder gets transferred to surrounding regions. Thus, to reduce the porosity defects in LPBF processing of copper and refractory metals, higher energy densities are required [49,55,110,111].

7.2. *Experimental data*

To assess the predictive capability of the constructed PV maps, experimental data were gathered from the literature for each corresponding material in Fig. 7. A summary of the selected experimental data is presented in Table 3. A one-to-one comparison of generated maps and available experimental data is only possible if the laser spot size, laser wavelength, and powder particle size, are similar in both scenarios. The PV maps are generated using a laser spot size of ~ 100 μm with a wavelength of ~ 1070 nm. Similar laser systems were used in the

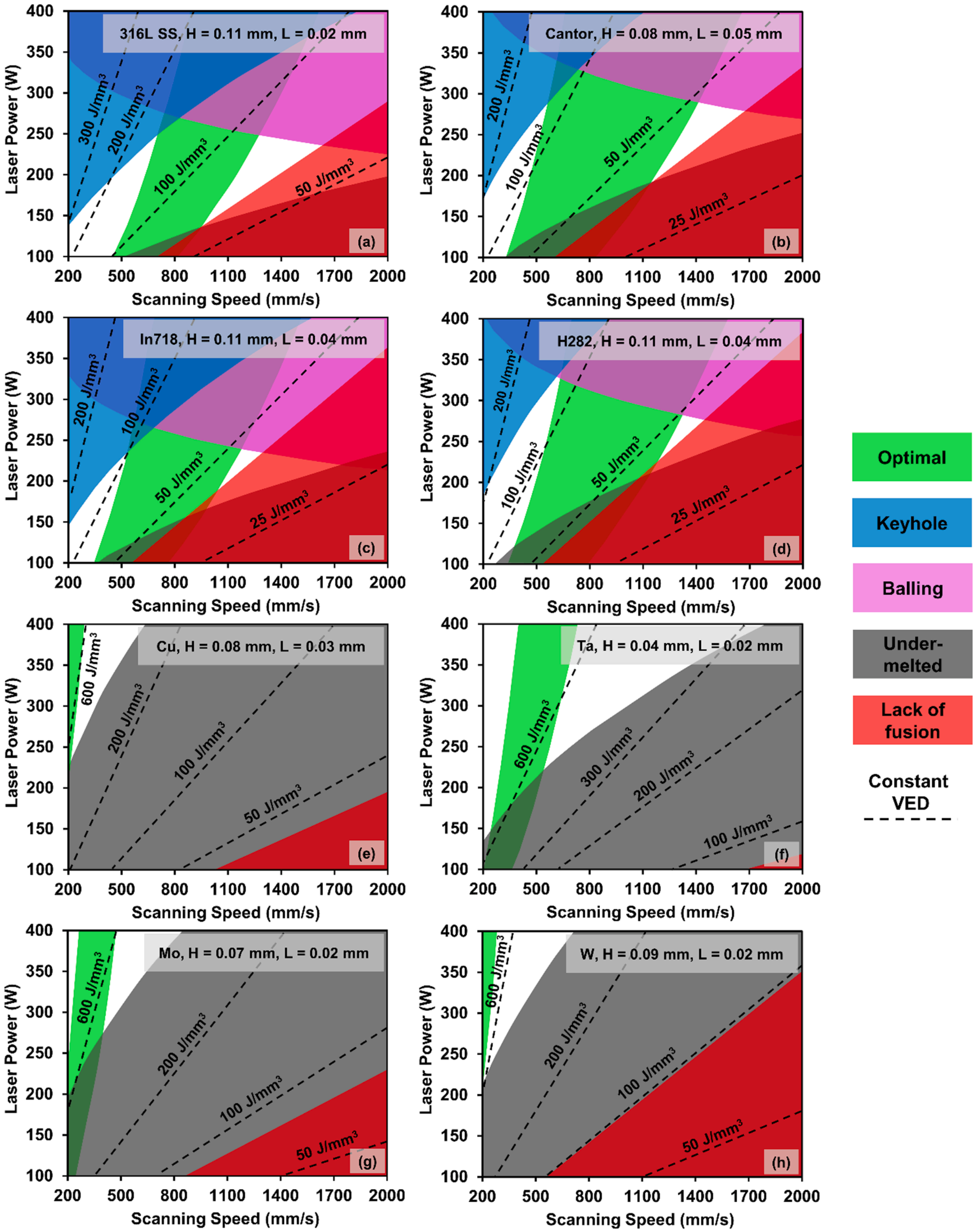

Fig. 7. Predictive PV processing maps for (a) 316L stainless steel (b) Cantor alloy (c) Inconel 718 (d) Haynes 282 (e) Copper (f) Tantalum (g) Molybdenum, and (h) Tungsten generated using analytical models.

collected experimental data. Moreover, similar hatch spacing, layer thickness and *P-V* range was used in the reported experimental data. Overall, the experimental data agrees with the analytical PV processing maps.

There are a few instances where the experimental data does not agree with the analytical predictions. For those cases, the experimental data is in fact in close proximity to the predicted region in the PV map. For every material under consideration in this work, the optimal parameter set that is experimentally obtained, is accurately predicted by the PV processing maps. This agreement highlights the predictive capability of analytical PV maps in guiding users to find the desirable processing parameters of materials under development for LPBF in a cost effective and timely manner.

Table 2
Selected analytical equations and corresponding thresholds used to construct PV maps.

Region	Equation(s)	Threshold
Lack of fusion	$Width(w) = \sqrt{\frac{8\alpha P}{\pi e \rho C_p V(T_m - T_o)}}$ $Depth(d) = \sqrt{\frac{2\alpha P}{\pi e \rho C_p V(T_m - T_o)}}$	$\left(\frac{H}{w}\right)^2 + \left(\frac{L}{d}\right)^2 \leq 1$
Balling	$Depth(d) = \frac{r_l}{\sqrt{p}}\begin{bmatrix} 0.008 - 0.0048B - 0.047p - 0.099Bp \\ +plnp(0.32 + 0.015B) + lnB(0.0056 - 0.89p + 0.29plnp) \end{bmatrix}$ $Length(l) = \frac{r_l}{p^2}\begin{bmatrix} 0.0053 - 0.21pB + 1.3p^2 - (0.11 + 0.17B)p^2 lnp \\ +B(-0.0062 + 0.23p + 0.75p^2) \end{bmatrix}$ $Width(w) = \frac{r_l}{Bp^3}\begin{bmatrix} 0.0021 - 0.047p + 0.34p^2 - 1.9p^3 - 0.33p^4 + \\ B(0.00066 - 0.007p - 0.00059p^2 + 2.8p^3 - 0.12p^4) + \\ B^2(-0.0007 + 0.015p - 0.12p^2 + 0.59p^3 - 0.023p^4) \\ +B^3(0.00001 - 0.00022p + 0.002p^2 - 0.0085p^3 + 0.0014p^4) \end{bmatrix}$	$\frac{l}{w} < 3.8$
Keyhole	$\frac{\Delta H}{H_s} = \frac{\alpha P}{\pi H_s \sqrt{D_t V {r_l}^3}} = \frac{\alpha P}{\pi \rho C_p T_m \sqrt{D_t V {r_l}^3}}$	$\frac{\Delta H}{H_s} < \frac{\pi T_b}{T_m}$
Under-melted	$\frac{\Delta H}{H_s} = \frac{\alpha P}{\pi H_s \sqrt{D_t V {r_l}^3}} = \frac{\alpha P}{\pi \rho C_p T_m \sqrt{D_t V {r_l}^3}}$	$\frac{T_m \Delta H}{H_s} < T_m$
Optimal	$E_{dim} = \frac{C_p P}{K V^2 H}$	$61 \leq E_{dim} \leq 146$

Table 3
A summary of reported experimental data in the literature pertaining to PV maps in Fig. 7.

Material	Power (W)	Velocity (mm/s)	Hatch Spacing (mm)	Layer Thickness (mm)	VED (J/mm3)	Reported Region	Predicted by PV maps	Ref.
SS316	120	600	0.11	0.02	91	LOF	No	Agrawal et al. [41]
	200	800			114	Optimal	Yes	
	200	1300			70	LOF	No	
	280	600			212	Keyhole	Yes	
Cantor Alloy	200	750	0.08	0.05	63	Optimal	Yes	Dovgyy et al. [112]
	200	430			110	Keyhole	No	
Inconel 718	100	1000	0.11	0.04	23	Under-melted	Yes	Scime and Beuth [79]
	250	200			284	Keyhole	Yes	
	250	800			71	Optimal	Yes	
	300	1400			49	Balling	Yes	
Haynes 282	200	1500	0.11	0.04	30	LOF	Yes	Islam et al. [113]
	250	975			58	Optimal	Yes	
	350	500			159	Keyhole	Yes	
Cu	150	600	0.08	0.03	104	LOF	Yes	Yan et al. [48]
	250	400			260	LOF	No	
	350	1000			146	Under-melted	Yes	
Ta	370	550	0.04	0.02	841	Optimal	Yes	Livescu et al. [49]
	370	770			601	LOF	No	
Mo	250	400	0.07	0.02	446	Keyhole	No	Higashi and Ozaki [111]
	250	1000			179	LOF	Yes	
	350	400			625	Optimal	Yes	
W	200	400	0.09	0.02	278	LOF	Yes	Tan et al. [114]
	200	300			370	LOF	Yes	
	200	200			556	LOF	Yes	

8. Summary and outlook

The present work provides a framework for designing processing parameters in the LPBF process based on a comprehensive review of available analytical models. The models used are mainly considered for melt pool geometry and the occurrence of porosity defects. These models are applied to a wide-ranging metals and alloys processed by LPBF to quickly generate and predict the processing maps. The processing maps provide information about the various types of porosity defects at different power-velocity combinations. The predictive capability of the processing maps is compared with existing experimental data from across the literature. Predictions can be improved significantly by improving the melt pool geometry estimates. Although these maps are approximations and some of the calculations rely on hard to acquire parameters such as laser absorption coefficient, analytical based PV maps can be computed quickly to provide an estimation for the processing window. This approach can be easily coupled with HT experiments or *in situ* monitoring to improve the predictability and verify the LPBF processing bounds.

The following summary captures the approach to generate predictive LPBF processing maps and fabricate dense LPBF components:

- Materials properties and process parameters are the key factors influencing the laser-metal interaction and melt pool geometry.
- Control over the melt pool geometry is essential to mitigate and minimize porosity defects. Melt pool geometry including width, depth, and length can be quickly estimated using the Rosenthal and Egar-Tsai based analytical models for various processing conditions.
- Defect criterion for different porosity defects such as lack of fusion, balling, and keyholing, coupled with melt pool geometry can be used

to estimate the processing bounds and generate predictive processing maps.

- Dimensionless numbers and design parameters help in correlating the processing parameters and reducing the dimensionality of the input parameter processing space.
- Existing analytical solutions for LPBF process can be combined to construct processing maps with excellent predictive capability for optimal or defective regions.

CRediT authorship contribution statement

Ankur K. Agrawal: Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Behzad Rankouhi:** Investigation, Data curation, Writing – review & editing. **Dan J. Thoma:** Conceptualization, Methodology, Supervision, Resources, Funding acquisition, Project administration, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: This report was prepared as an account of work sponsored by an agency of the United States Government. Neither the United States Government nor any agency thereof, nor any of their employees, makes any warranty, express or implied, or assumes any legal liability or responsibility for the accuracy, completeness, or usefulness of any information, apparatus, product, or process disclosed, or represents that its use would not infringe privately owned rights. Reference herein to any specific commercial product, process, or service by trade name, trademark, manufacturer, or otherwise does not necessarily constitute or imply its endorsement, recommendation, or favoring by the United States Government or any agency thereof. The views and opinions of authors expressed herein do not necessarily state or reflect those of the United States Government or any agency thereof.

Data availability

All data included in text

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